

Removing the Finite-Support Hypothesis: Literature Resolution of the Question in arXiv:math/0004146

Run fa_banach_001 — literature-status note

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Status. `literature_already_answered`. This note identifies an exact later affirmative solution. It does not claim a new theorem.

The source question

Let $(\mathcal{M}, \tau)$ be a semifinite von Neumann algebra and let E be an order-continuous symmetric Banach function space with the Fatou property. Randrianantoanina’s Proposition 2.6 in the arXiv draft (Proposition 3.4 in the revised journal paper) considers a normalized sequence (x_n) in $E(\mathcal{M}, \tau)$ under the additional hypothesis

$$\text{there is } e \in \mathcal{M} \text{ with } e^2 = e, \quad \tau(e) < \infty, \quad ex_n = x_n \quad (n \geq 1). \quad (\text{F})$$

It proves the following alternative: either there is $C > 0$ such that

$$\int_0^1 \left\| \sum_{i=1}^n r_i(t) a_i x_i \right\|_{E(\mathcal{M}, \tau)} dt \geq C \left(\sum_{i=1}^n |a_i|^2 \right)^{1/2} \quad (\text{R})$$

for every n and every scalar choice, or (x_n) has a subsequence equivalent to a disjointly supported sequence in E .

Immediately afterward the paper says: “We do not know if condition (ii) can be removed” [1, Remarks 3.5]. Condition (ii) is exactly (F). The 2003 journal revision retains this sentence, so this was not merely an obsolete line in the 2000 arXiv draft.

Exact later answer

Huang and Sukochev prove the noncommutative Lindenstrauss–Tzafriri dichotomy at the level of general operator bimodules [2]. Their published abstract states the result in the following form.

Theorem (Huang–Sukochev). *Let $p > 0$ and let $\mathcal{E} \subset L_p(\mathcal{M}, \tau) + \mathcal{M}$ be a noncommutative quasi-Banach $\mathcal{M}$ -bimodule with order-continuous quasi-norm. For a bounded (in particular, normalized) sequence (x_i) in $\mathcal{E}$, either there is a constant $c > 0$ for which*

$$\int_0^1 \left\| \sum_{i=1}^n r_i(t) a_i x_i \right\|_{\mathcal{E}} dt \geq c \left(\sum_{i=1}^n |a_i|^2 \right)^{1/2} \quad (n = 1, 2, \dots),$$

for every scalar choice, or (x_i) has a subsequence equivalent to a sequence of disjoint elements.

Most decisively for provenance, the same abstract explicitly says that this answers the question of Randrianantoanina in the 2003 paper [2].

Why this removes precisely (F)

Take $\mathcal{E} = E(\mathcal{M}, \tau)$. It is a Banach $\mathcal{M}$ -bimodule, and order continuity of E gives order continuity of its norm. As used already inside the source proof, the symmetric space embeds continuously into $L_1(\mathcal{M}, \tau) + \mathcal{M}$; hence the later theorem applies with $p = 1$. A normalized sequence is bounded, and no common finite-trace support projection occurs in the later hypotheses.

The first later alternative is word-for-word (R). In the second alternative, disjoint operator elements have orthogonal left and right supports. The standard singular-value disjointification used in the source’s Proposition 2.4 identifies such a sequence, up to equivalence, with a disjointly supported sequence in the underlying symmetric function space E . Thus the second alternative is also the source conclusion.

Consequently, condition (F) can be removed, affirmatively and without adding the type-2 assumption used elsewhere in the 2003 paper. In fact, the later theorem is stronger: it works for order-continuous quasi-Banach operator bimodules, not only symmetric Banach spaces with the Fatou property.

Scope and verification trail

This packet addresses only Remarks 3.5/its arXiv-draft counterpart. It does not reclassify the paper’s separate embedding theorems for noncommutative Lorentz spaces. The included 2003 publisher PDF verifies the final wording of the source question. The DOI metadata and the complete published abstract of [2] are included as JSON snapshots; the abstract both states the unrestricted alternative and names the source question as an application.

References

- [1] N. Randrianantoanina, *Embeddings of ℓ_p into non-commutative spaces*, J. Aust. Math. Soc. **74** (2003), 331–350; arXiv:math/0004146.
- [2] J. Huang and F. Sukochev, *Bounded sequences in noncommutative bimodules of τ -measurable operators*, Proc. Amer. Math. Soc. **154** (2026), no. 2, 793–806, doi:10.1090/proc/17465 (published online December 1, 2025).